

Research Paper 13 Summary

A Unified Approach to Interpreting Model Predictions (SHAP)

Problem Statement: AI-Based Detection and Classification of Industrial Fires (PS26162)

Original Paper: Lundberg & Lee (2017), *NeurIPS*

1 Basic Information

- **Paper Title:** A Unified Approach to Interpreting Model Predictions
- **Authors:** Scott M. Lundberg and Su-In Lee (2017)
- **Conference:** *Advances in Neural Information Processing Systems (NeurIPS)*
- **Core Contribution:** Introduced the **SHAP (SHapley Additive exPlanations)** framework, the global gold standard for Explainable AI (XAI).

2 What is this Paper About? (In Simple Words)

Modern machine learning models (like XGBoost and Random Forest) are powerful, but they operate as “black boxes”—they output a prediction, but humans cannot easily see *why* the model made that choice.

- **The Black Box Dilemma:** If an AI model alerts a disaster management team that a thermal spot is a “Critical Industrial Fire”, the operator will ask: “*Why did the AI say this? Is it a false alarm?*”
- **The SHAP Solution:** Based on Nobel Prize-winning cooperative game theory (Shapley values), SHAP calculates the exact positive or negative contribution of every single feature toward a specific prediction.
- **Transparent Math:** For every alert, the sum of all SHAP values equals the final prediction score, showing exactly which clues pushed the model toward its conclusion.

3 Key Concept: Feature Contribution Breakdown

For any given satellite thermal alert, SHAP provides a transparent calculation:

$$\text{Final Prediction Score} = \text{Base Value} + \sum_{i=1}^M \phi_i \quad (1)$$

Where ϕ_i is the SHAP contribution of feature i .

4 How This Helps Our Project (PS26162)

4.1 1. Powering the Explainable AI (XAI) Panel in the GIS Dashboard

When a user clicks on any fire alert icon on our GIS map overlay, our system displays a transparent, human-readable explanation generated by SHAP:

Alert: Classified as *Industrial Flare* (94% Confidence)
Why?

- +42% contribution: Located 60 m from an OSM Oil Refinery polygon.
- +31% contribution: Heat detected 18 times in the last 30 days.
- +16% contribution: Flame temperature calculated at 1380 K.
- −5% contribution: Located near agricultural land cover.

4.2 2. Building Trust with Emergency Responders

Industrial plant safety managers and government disaster response officials cannot take action based on unexplained AI scores. SHAP provides the transparency required for real-world deployment and safety audits.

4.3 3. Global Model Debugging and Feature Verification

Using SHAP summary plots during model development allows our team to ensure the AI is learning true physical rules (e.g., thermal stability and factory proximity) rather than overfitting to spurious dataset correlations.